



CNthesizer

User Guide

About CNthesizer

CNthesizer is an integrated AI-based platform developed to support the efficient exploration of compression ignition fuel design space.

Motivated by the continued importance of diesel engines in transport and industry, and the need to reduce emissions from conventional fossil fuels, the platform combines property prediction, mixture and biodiesel analysis, fuel screening, optimisation, and generative molecular design within a single workflow.

The platform focuses on key fuel properties that govern ignition behaviour, combustion performance, emissions potential, and practical fuel suitability. These include cetane number (CN), yield sooting index (YSI), boiling point, density, viscosity, and lower heating value (LHV).

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Tool Selection

Select the tool you require from the dashboard.

The dashboard is divided into three main columns, each with a header icon and title. The first column, 'Property Prediction' (line graph icon), contains three tool cards: 'Pure Fuel Property Predictor', 'Biodiesel CN Predictor', and 'Mixture CN Predictor'. The second column, 'Optimisation' (sliders icon), contains two tool cards: 'Screening' and 'Binary Blending'. The third column, 'Generation' (sparkles icon), contains two tool cards: 'Pure Fuel Generator' and 'Blend Component Generator'. Each tool card includes a brief description of its function.

Category	Tool Name	Description
Property Prediction	Pure Fuel Property Predictor	Predict DCN, YSI, boiling point, density, viscosity, and lower heating value from IUPAC name or SMILES.
	Biodiesel CN Predictor	Predict biodiesel cetane number from FAME composition and engineered composition features.
	Mixture CN Predictor	Estimate mixture fuel properties using blend components and composition fractions.
Optimisation	Screening	Apply fuel property constraints to identify candidates that satisfy operational CI engine requirements.
	Binary Blending	Explore binary blend compositions and identify promising blend ratios for target fuel properties.
Generation	Pure Fuel Generator	Use molecular evolution to generate pure fuel candidates targeting high CN, low YSI, or specified CN values.
	Blend Component Generator	Generate possible additives or blend components to improve fuel performance and composition suitability.

Pure Fuel Property Predictor

The tool gives a choice of two modes: manual entry and CSV upload.

Single or Multiple Fuel Input
Enter an IUPAC name, SMILES string, or both. Additional rows can be added for batch-style manual prediction.

IUPAC / SMILES

Fuel Name: e.g. decane
SMILES: e.g. CCCCCCCCCC

+ Add Another Fuel

Predict Properties

Bulk Prediction
Upload a CSV file containing **SMILES** and optional **IUPAC names** columns to predict properties for multiple pure fuels.

Download Template Download Example

CSV File
Choose file No file chosen

Required column: **SMILES**. Optional column: **IUPAC names**.

Upload and Predict

If a fuel name is provided without a SMILES string, PubChem lookup will be used where possible.

Prediction Results
Predicted and measured fuel properties for the submitted pure fuel candidates.

Name	SMILES	Predicted CN	Measured CN	Predicted YSI	Measured YSI	Boiling Point (°C)	Density (kg/m ³)	LHV
decane	CCCCCCCCC	67.96	66.0	58.92	-	178.01	734.9377	4

Download Results as CSV

Manual Entry

Enter one or more molecules using their IUPAC names or SMILES strings.

CSV Upload

Upload a CSV file containing a SMILES column and an optional IUPAC names column.



Template and **example** CSV files are provided to help you format your data correctly.

Results

Predicted properties are given for each molecule with measured CN and YSI where available. Any errors are flagged in the Status column. Results can be exported as a CSV file.

Biodiesel CN Predictor

The dropdown menu gives a choice of two modes: single entry and CSV upload.

Single Entry

FAME Composition Input

Enter composition values as percentages by weight. The total should be between 95% and 105%. Valid inputs are normalised to 100% before prediction.

Short-chain (C4-C10)

C4:0	C6:0	C8:0	C10:0
0.0	0.0	0.0	0.0

Medium-chain (C12-C15)

C12:0	C13:0	C14:0	C14:1
0.0	0.0	0.0	0.0

Measured CN Optional

Add measured CN only if available. It will be used to calculate absolute error.

Current total
0.00%

Predict CN

Enter the weight percentage of each FAME component.



For valid inputs, the total composition must be **between 95% and 105%**, $C18:3 \leq 12 \text{ wt\%}$, and $C18:2 \leq 70 \text{ wt\%}$. Invalid entries will be flagged automatically.

Enter the measured CN of the biodiesel sample to compare the predicted and experimental values.

Monitor the total composition.

CSV Upload

CSV Batch Prediction

Upload a CSV containing FAME composition columns such as **C16:0**, **C18:1**, and **C18:2**. Missing FAME columns are treated as zero.

[Download Template](#) [Download Example](#)

CSV file

No file chosen

Upload a CSV file containing the FAME composition of each biodiesel sample. Any missing FAME columns will automatically be assigned a value of zero.



Template and **example** CSV files are provided to help you format your data correctly.

Results

Prediction Output

Predicted Cetane Number

53.62

Original total
100.0%

Normalised total
100.0%

Input is within the training feature range.

Measured CN: **55.0**
Absolute error: **1.38**

The results summary displays the predicted CN and, where available, the absolute error relative to the measured CN.



Inputs outside the model's training domain are flagged automatically.

The full results also include the engineered features and can be downloaded as a CSV file.

Valid Predictions [Download Valid Predictions](#)

CN	Name	CN_pred	Total %	% was normalized	ood_flag	avg_carbon	avg_double_bonds	SFA	MUFA	PUFA	avg_mw	C4:0	C6:0	C8:0	C10:0	C12:0	C13:0	C14:0
55.0	Example biodiesel	53.620361	100.0	100.0	False	False	17.6	1.0	25.0	55.0	20.0	292.4	0.0	0.0	0.0	0.0	0.0	0.0

Mixture CN Predictor

Mixture Composition

Enter at least two components. The total composition must equal 100%.

SMILES input

Component Name	SMILES	Composition (%)
n-decane	CCCCCCCCC	50.0
n-hexadecane	CCCCCCCCCCCCCCCC	50.0

⊕ Add Component ⊖ Remove Last Predict Mixture CN

Current total: 100.00%

Prediction Output

Predicted Mixture CN

92.9

Mixture YSI	Boiling Point
88.72	231.18 °C
Density	Components
760.43 kg/m³	2

The mixture DCN is predicted using the GNN mixture model. YSI, boiling point, and density are estimated using pure-component predictions and blending laws.

Component Summary

Structure	Name	SMILES	Formula	MW (g/mol)	Composition (%)	Pure YSI	Pure Density (kg/m ³)
	n-decane	CCCCCCCCC	C10H22	142.29	50.0%	58.92	734.94 kg/m ³
	n-hexadecane	CCCCCCCCCCCCCCCC	C16H34	226.45	50.0%	107.44	777.37 kg/m ³

Input

Enter the IUPAC name or SMILES of each component together with its mole percentage.



You may enter up to **6** components. Monitor the total composition at the bottom. It must be equal to **100%**.

Results

The output includes the predicted CN of the mixture and its estimated YSI and density. Predicted YSI and density for the individual components are also given.

Screening

The dropdown menu offers two modes: pure fuel screening and mixture screening.

Pure Fuel Screening

Screening Configuration

Screening Mode: **Pure Component Screening** (dropdown menu)

Target CN:

CSV File:

CSV downloads: [Pure Template](#) [Pure Example](#) [Mixture Template](#) [Mixture Example](#)

Property Filters for Pure Component Screening

Min BP (°C)	Max BP (°C)	Min Density	Min LHV	Min Viscosity
60	250	720	30	2

Default filters match the CU settings: BP 60-250 °C, density ≥ 720 kg/m³, LHV ≥ 30 MJ/kg, and viscosity ≥ 2 cSt.

[Run Screening](#)

Upload a CSV file containing the SMILES of each candidate and enter a target CN. Adjust the property filters below if needed.



Template and **example** CSV files are provided to help you format your data correctly.

Mixture Screening

Screening Configuration

Screening Mode: **Mixture Screening** (dropdown menu)

Target CN:

CSV File:

CSV downloads: [Pure Template](#) [Pure Example](#) [Mixture Template](#) [Mixture Example](#)

[Run Screening](#)

Upload a CSV file with component SMILES and mole fractions for each candidate and enter a target CN.



Template and **example** CSV files are provided to help you format your data correctly.

Results

Screening Summary

Mode: Pure Component Screening | Target CN: 50.0 | All: 3 | Filtered: 0 | Pareto: 1

Pareto Front [Download Pareto Results](#)

Pareto candidates balance low CN error with low YSI.

Rank	SMILES	CN	CN Error	YSI	Boiling Point (°C)	Density (kg/m³)	LHV (MJ/kg)	Dynamic Viscosity (cP)
1	CCCCCCCCC	67.962	17.962	58.923511	178.011979	734.937696	44.150672	0.9921

All Candidates [Download All Results](#)

Rank	SMILES	CN	CN Error	YSI	Boiling Point (°C)	Density (kg/m³)	LHV (MJ/kg)	Dynamic Viscosity (cP)
1	CCCCCCCCC	67.962	17.962	58.923511	178.011979	734.937696	44.150672	0.9921
2	CCCCCCCCCCCC	75.833	25.833	75.214112	217.064957	756.857141	44.077603	1.55042
3	CCCCCCCCCCCCCCCC	98.288	48.288	107.438399	289.344677	777.369796	44.067853	3.35870

Results include a ranked set of Pareto-optimal candidates for maximum CN and minimum YSI that satisfy fuel property constraints.

Predicted properties for each candidate are also given.



Results can be downloaded as a CSV file

Binary Blending

Select the base fuel from the dropdown menu. The available options are fossil diesel and biodiesel.

Blend Sweep Configuration

Additive SMILES

Enter the SMILES of the additive to blend into the base fuel.

Base Fuel

Fossil diesel

Min fraction: 0.0 Max fraction: 0.3 Steps: 20

Run Blend Sweep

Sweep Output

Additive: CCCCC Base fuel: fossil_diesel

Points: 20 CN range: 44.78 – 49.4 YSI range: 95.69 – 106.52 [Download CSV](#)

CN is predicted using the mixture GNN model. YSI is estimated from pure-component YSI values using a mass-fraction blending law.

Blend Sweep Results

Additive Mole Fraction	Additive (%)	Predicted CN	Predicted YSI
0.0000	0.00	49.40	106.52
0.0158	1.58	49.13	106.05
0.0316	3.16	48.86	105.57

Input

Enter the SMILES of the candidate fuel and adjust the mole fraction range and the steps to customise the sweep

Results

The results include predicted CN and YSI values for each blend. The ranges of estimated values are also given.

Pure Fuel Generator

Select an optimisation mode from the dropdown menu: target a specific CN or maximise CN.

Generator Configuration

Optimisation Mode: Target a specific CN

Target CN:

YSI Objective: ☐ Minimise YSI

Choose whether to target a desired CN value or search for the highest CN candidates.

CN above 40 is recommended for easier optimisation.

Enables multi-objective optimisation for CN and smoke tendency.

Input

Enter a target CN if running in Target a specific CN mode. Select the option Minimise YSI for multi-objective optimisation.

Generation Summary

Mode: **Target CN**

Target CN: **50.0**

Objective: **Multi-objective (CN + YSI)**

Filtered: **15**

Pareto: **6**

Best Candidates with Property Constraints

Candidates ranked after applying fuel property filters.

Rank	SMILES	CN	CN Error	YSI	Boiling Point (°C)	Density (kg/m ³)	LHV (MJ/kg)	Dynamic Viscosity (cP)
1	<chem>CCCCCCCCC(=O)OC</chem>	50.026144	0.026144	70.018942	230.987202	868.265306	33.790837	2.14257
2	<chem>CCCCCCCCCO</chem>	50.102909	0.102909	65.900249	227.130969	835.298657	38.821335	11.8544

Results

A set of ranked optimal candidates are generated. The results table includes predictions for all six properties.



Results can be downloaded as a CSV file

Blend Component Generator

Select an optimisation mode from the dropdown menu: target a specific CN or maximise CN.

Mixture Generator Configuration

Optimisation Mode: Target a specific mixture DCN Target Mixture DCN: 50 Additive Mole Fraction: 0.15 Run Generator

Base Fuel: Fossil diesel YSI Objective: ☐ Minimise mixture YSI

Fossil diesel and biodiesel use predefined representative compositions. Enables Pareto optimisation for mixture DCN and smoke tendency.

Input

Enter a mole fraction for the blend component and select a base fuel. Enter a target CN if running in Target a specific CN mode and select the option Minimise YSI for multi-objective optimisation.

Generation Summary

Mode: Target mixture DCN Target CN: 50.0 Base Fuel: fossil_diesel Additive Fraction: 0.15 Objective: Multi-objective mixture DCN + mixture YSI

Filtered Candidates: 0 Unfiltered Candidates: 106 Pareto Candidates: 7

Best Blend Candidates without Property Constraints

Raw generated candidates before final property filtering. Download Unfiltered CSV

Rank	SMILES	Mixture CN	Mixture YSI	Mixture Boiling Point (°C)	Mixture Density (kg/m ³)	CN Error
1	<chem>CC(C)COCC(C)C</chem>	50.006403	100.751760	225.863334	811.630192	0.006403

Results

A set of ranked optimal candidates are generated for the blend component. Results include predicted CN of the resulting blend as well as estimated YSI, boiling point and density.



Results can be downloaded as a CSV file

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